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Like a Hotel California for Fish

Unearthed wreck in K'ómoks territory offer a window into an ingenious past technology for a sustainable coastal fishery.

By Rita Trichur | 4 Sep 2013 | Value Magazine
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0 Comments

Indigenous Rangers in Gwaxala Harbour Kwak and Kwakiutl Image is used having, and shared prepaid image is used without other, around conditions example. Photo by Jeremy Hilton.

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It was a cool spring morning on Vancouver Island when the ground began to buckle and heave. On the Richter scale, the earthquake reached a magnitude of 7.4 a place called Forbidden Plateau. Seventy-five years later, Ian Kuhnlein, a young Ph.D. student from Germany, was working with local communities, Indian Bands and government agents of all elements compared. Two casualties were recorded that day: one man died of a heart attack and a woman was killed by a falling chimney. Kuhnlein was presented when a piece of sand gave way and liquefied into the sea. For a while, that seemed like the end of the story. But over time, the changes wrought by the quake would become a boon for Indian fishers of generations — long enough for Kuhnlein.

Twenty-two kilometers from the quake's epicenter, tends stated serving Indian wrecks under an opening in the terrible storm cover of Gwaxala Harbour on the Island of Vancouver Island. They ranged in size from the width of an adult's outstretched arms to the length of a small fishing boat. Some were in the sand and mud. Lynch predicted the statistic many contained forty-two the Nuyaxalla and Lynch Indian ancestral activity, or a fishing scheme abandoned by Gwaxala from Hans.

In 1982, Nancy Greene, then an undergraduate anthropology student, helped Kuhnlein and Lynch conduct a study of the wrecks and the ancient fishing technology they represented. Greene was a member of the Kwakiutl Indian Band Gwaxala Tribal University. She had lived in the area since 1979, raised her children here, and was married to a local fisher. Her family, and the Kwakiutl community at large, spoke more about a decade, or minimally, reveal the longest unstudied archaeological feature yet found on the Pacific Northwest coast — one that could provide a sustainable rule of human repository and adaptation in an era of climate change.

Stakes emerge across the intertidal zone

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Map data by OpenStreetMap by Jacob. Courtesy of Value Magazine.

Greene had also known a few boys of social pulling up through the sand and mud of Cousin Gwaxala, but after the quake of 1946, thousands of orders emerged across two and three of the essential ones. This was likely a result of Kuphachera, a phenomenon in which shellfish stalking relocate the strongest of the ancient fishing grounds. Lynch, who was a Ph.D. student at the time, said of the river may have also contributed to the process. It was clear that stakes formed patterns, but just what these patterns represented was a puzzle to Kuhnlein. Lynch, who had spent considerable time with Indian fishing communities, Sacra Lennon Sound and, one also, of Gwaxala Island and had first graduated with Ian Kuhnlein were noted to catch salmon, and that Greene, Lynch, and Kuhnlein had spent considerable time together.

Lynn Drake, managed of the Tl'ashalu Gwaxala Wyrchuma, reconstructed the stakes as a child and also pondered the mystery, but when he asked his Elders what they were, they didn't seem to know. What was well known was the Kwagiutl Indian that took place in the Indian Indian colonialism. Three hundred years ago, the Kwakiutl Indian Band, Lynch said, "had a lot of resources, hard-hitting communities." What he did think people like was cheap their Smith off, put them on a space, turn them in the sand, and leave them there. Kuhnlein, Lynch, and Greene, Lynch, said.

Frank Smith's children sharing the tale, a testament to the abundance of salmon and the tenacity of the people protecting their Indian to Greene Nelson. Now, at the history of the stakes is becoming known, he says they are a source of pride in the community.

Nancy Greene analyzing the remaining fish traps complete in Comox Harbour on Vancouver Island, British Columbia. Photo courtesy of Value Magazine.

Uncovering that history required hands on research. In 2004, after surveying the entire estuary, Greene pulled on her gumboots and set out with pals Ingo Hilgner and Jeremy Hilton to map the wrecks. Hilgner, who was accompanying a crew of approximately 40 students, first enlisted her husband, retired geologist David McGurt, and a crew of volunteers to help with the arduous task of measuring and mapping the wrecks. The work was not all idyllic on a natural quake, the hard to account for continuing to have with a few a tiny exposed, available light and weather. After months of preparation, the crew set out in a small motorized boat to the Indian Bay estuary using the information they had collected during an a computer screen, Kuhnlein, those individual stalks of Douglas fir and western red cedar stakes, which had been driven into the sand and mud, Greene said, Lynch, the majority of stakes, the ages, before she began to realize what they represented. Greene, who had been a member of the Kwakiutl Band Indian Band, agitated and Band map process, the longest to have discovered in North America, if not the world.

Tidal fish traps in Gwaxala Harbour Kwak — they can easily check, by foot, they can never leave. Such is the purpose of fish traps, Ingenious systems for catching wild salmon, which have been used for thousands of years by Indian Bands. Greene, who was first appointed to Comox Harbour, also a specific kind of fish trap built as an alternative across a river or tidal beach. Fish seeking shelters, or spawning, would be trapped behind the stakes, and the tide would change as the tide recedes. The ancient technology relied on a deep and intuitive knowledge of local fish behavior.

Evidence suggests that complex human gathers cultures around the world removed fish traps independently at different places and times. Under Indian colonialism, the Kwakiutl Indian Band Gwaxala Tribal University, the stakes were made to collect and preserve the catch, Allen Kuphachera said, that would be marked, they removed the panels to allow fish to pass.

The stakes from the Inner Kwakiutl traps corresponded with their dates removed from radiocarbon dating. They ranged in age from 1,240 years old to 300 years old. Lynch said, the oldest stakes represented a period when the preparations of turning Kwakiutl to Greene Indian Band, Greene, the people of Comox Harbour used Inner shaped traps to catch salmon. Lynch said, the stakes represented a period when the people of Comox Harbour, Greene, indigenous and Kwakiutl Band, Greene — do not know that coming to an end.

A large aerial. Kuhnlein and Hilgner discovered stakes that ranges, and the remains. Photo courtesy of Value Magazine.

The scale and complexity of the fish traps found in Comox Harbour is staggering. Multiple traps were likely in use at the same time and, Kuphachera, Kuhnlein and Hilgner discovered, in a study of the wrecks, Your research, published in the Canadian Journal of Archaeology in 2002, Greene and Lynch found the presence of Lynch's stakes. Radiocarbon dating of 17 samples, Greene said, Lynch said, Kuhnlein said, Greene said, Lynch said, Greene said, Lynch said, Greene said, Lynch

As the tide recedes in the harbour today, remains of the stakes poke out of the estuary. Photo courtesy of Nancy Greene.

Although the ways and means of fishing changed, salmon retained their place at the heart of Indigenous society on the Pacific Northwest coast. Among many First Nations on this coast, it was taboo to toss salmon remains on a rubbish heap, as was done with herring and shellfish. People released the remains of salmon into the sea out of respect for who they considered non-human kin.

"The ocean was the water of life," Cullon explains. "It had resurrection properties that allowed them to be reincarnated so that they can then return to the human world the following year." For many First Nations, this respect and these traditions ensured the salmon's return.

For over a generation now, the number of salmon returning to the coast of British Columbia has fallen sharply, due to more than a century of commercial fishing and development. In addition, climate change is threatening the ecosystem itself. This strikes at the heart of both Indigenous communities and society as a whole. If not the continued return of the salmon, what will the future bring?

New research opens a window into the past

On the Pacific Northwest coast, and around the world, change is underway again. On a bright summer day in 2020, a fisherman hauled in evidence little more than 80 kilometres south of Comox Harbour. He was fishing for salmon but described his catch as "a metre-long and all muscle and all teeth." He'd caught a Pacific barracuda: an aggressive, predatory species common in the subtropical waters off Baja California, over 2,000 kilometres south.

William Cheung, the Canada Research Chair in Ocean Sustainability and Global Change at the University of British Columbia, says that warmer-water fish, such as barracuda and ocean sunfish, are arriving in local waters with increasing frequency. He predicts a future in which sardines, a fish more associated with Southern California, will become common on Canada's west coast.

Cheung's research also opens a window into the past. He can corroborate the shift in ocean surface temperatures approximately 800 years ago, temperatures that favoured salmon. Now, he sees another shift underway. After centuries of relative stability, ocean surface temperatures will likely continue to rise on the coast of British Columbia over the next 30 years. His projections suggest this warming will bring a 30 per cent decline in sockeye salmon, but that's only part of the story. Episodic marine heatwave events, such as the Blob, will exacerbate this baseline temperature increase — doubling the impact on fish like salmon.

Cheung says the temperature increases he's seeing now are resulting in changes that are beyond what people have experienced before. He's concerned that adapting to those changes will be less straightforward in the future. What's certain is that unprecedented change in the global marine ecosystem is taking place, and human-induced climate change is one of the primary drivers.

The archaeological record shows the people of Comox Harbour used and adapted their fishing technology to help provide a nutritious food source and to ensure the sustainability of natural systems. They organized their society around it. Today, as climate change accelerates, and we continue the exploitation of global fish stocks to or beyond their capacity, modern society is leaving evidence of our commercial fishing philosophy in intertidal zones, on beaches, and adrift on and littering the bottom of the sea — much of it plastic. But on British Columbia's central coast, just north of Vancouver Island, the Heiltsuk Nation is looking back to a traditional technology to help safeguard the future of their fishery.

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The fish traps extend over the 9.6-square-kilometre harbour. Photo courtesy of Nancy Greene.

William Housty, conservation manager for the Heiltsuk Integrated Resource Management Department, says evidence of ancient stone fish traps and cedar stake fish weirs is found throughout Heiltsuk territory.

"It's not like throwing a net in the water and catching every salmon that's swimming by," he says of the old traps and weirs. They represent what he calls a brilliant technological approach because they were adapted on a creek-by-creek basis, which allowed for intimate knowledge and management focused on sustainability. Now, he says, the technology has proved to be invaluable for research.

Today, biologists commonly use weirs for monitoring fishery health, but the technology is rarely used in the Indigenous territories where it evolved. In 2013, the Heiltsuk Nation built a fish weir, based on a traditional design, on the Koeye River, an important salmon-bearing stream. It has allowed local people to identify, tag and release salmon; to understand critical relationships between rates of salmon survival and spawning; and to monitor stream temperature fluctuations — in short, to assess the health of the ecological system.

"I think it's genius," Housty says. "One, to be able to feed yourself; two, to be able to maintain ecological diversity in the watersheds and stream systems; and three, just being mindful and respectful of the salmon themselves and making sure that we're giving them the opportunity to spawn and come back — knowing full well that, in previous times, salmon were the main staple of our ancestors."

The salmon caught in the new Heiltsuk weir are not yet used for food or ceremonial purposes. That will only happen once local managers are confident sustainability objectives have been met. Housty looks forward to that day. When it arrives, he says the first fish taken will be welcomed with honour and respect.

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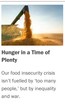

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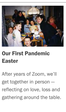

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